

deterministic decoding ($m = 0$)

*attacker
find one
witness*

JAILBREAK-EXISTS

$$\exists p : \text{leak}(p) = 1$$

NP-complete (Thm 1)

↓ *from $\exists$ to $\forall$ over all prompts*

SAFE = CERTIFY₀

$$\forall p : \text{leak}(p) = 0$$

co-NP-complete (Thm 2)

↓ *add $\exists d$ with $|d| \leq \beta$*

DEFEND

$$\exists d : \text{Adm}(d) \wedge \forall p : \text{leak}_d(p) = 0$$

Σ_2^P -complete (Thm 5)

*defender certify
all prompts meet
the leak bound*

*designer
choose a defense
for all prompts*

stochastic decoding ($m > 0$)

BREACH

$$\exists p, r : \text{leak}(p, r) = 1$$

NP-complete (Thm 1)

*LeakCount: #P-complete
exact $q(p)$: #P-hard (Thm 3)*

CERTIFY

$$\forall p : q(p) \leq \tau$$

co-NP^{PP}-complete (Thm 4)



STOCHASTIC DEFENSE DESIGN
not formalized in this paper